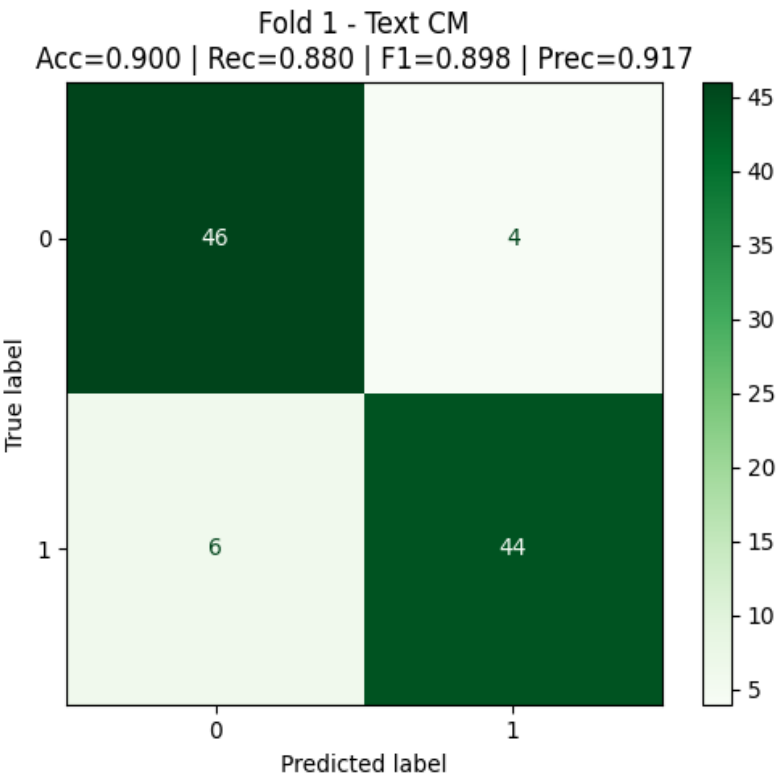


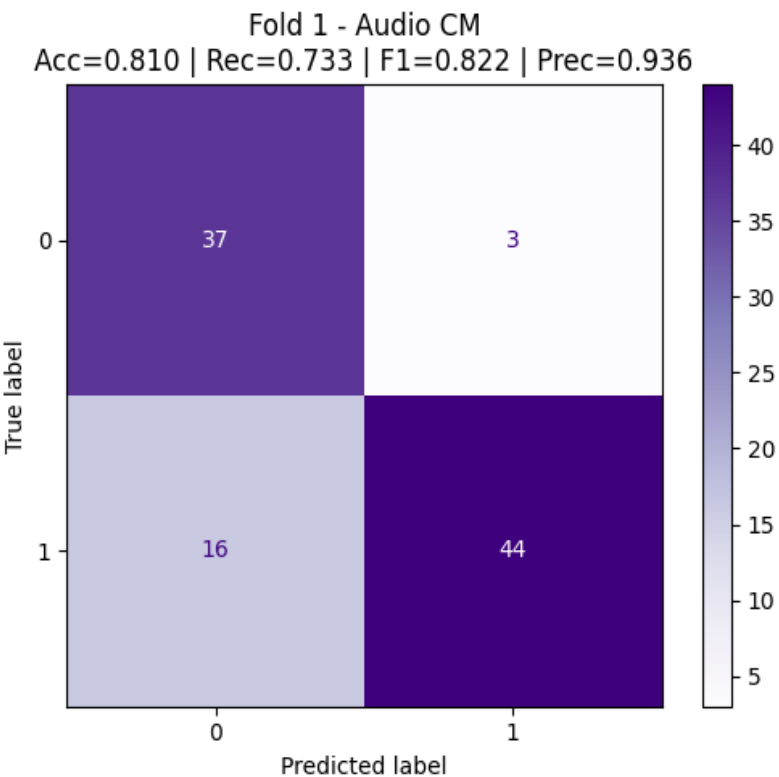
Multimodal Suicide Detection - Cross Validation Report

Fold 1 Results

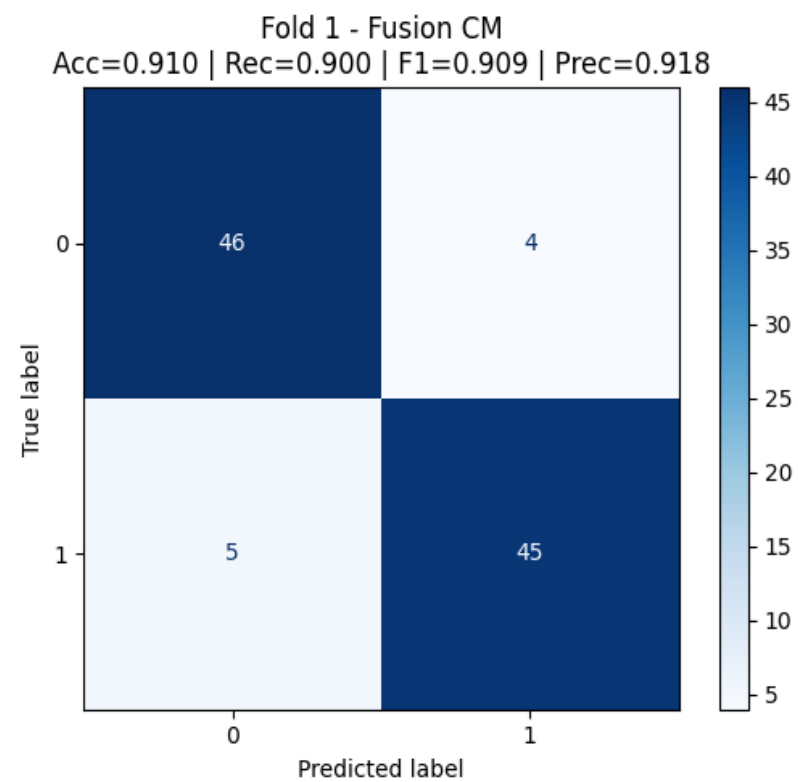
Text Confusion Matrix



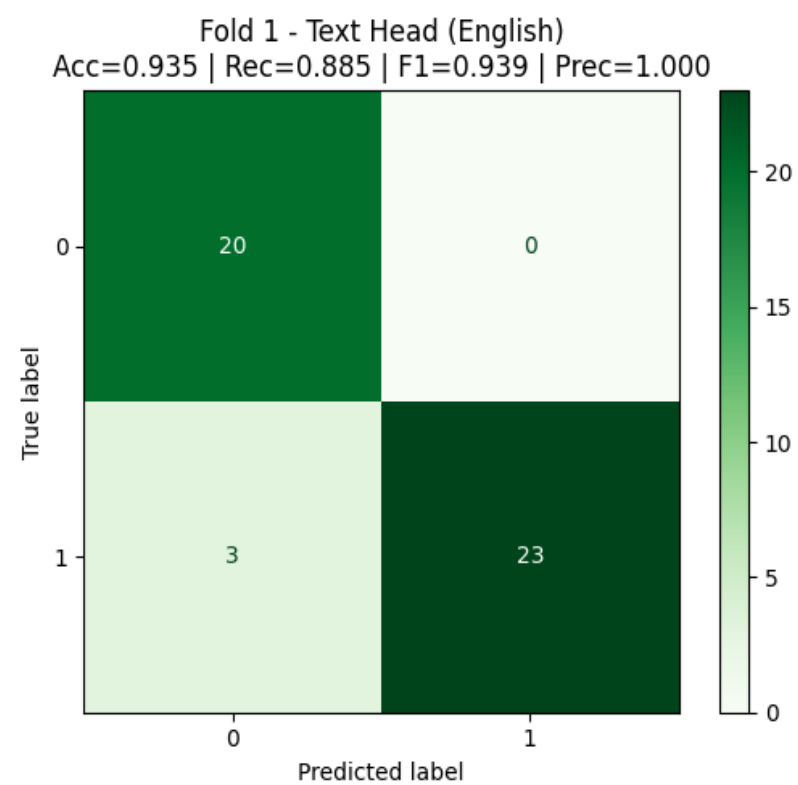
Audio Confusion Matrix



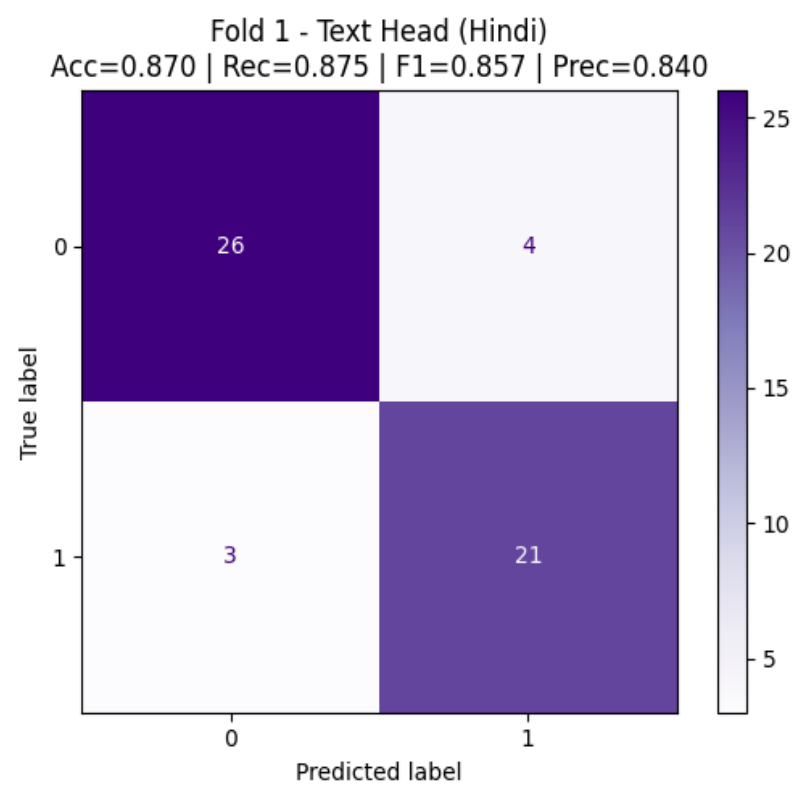
Fusion Confusion Matrix



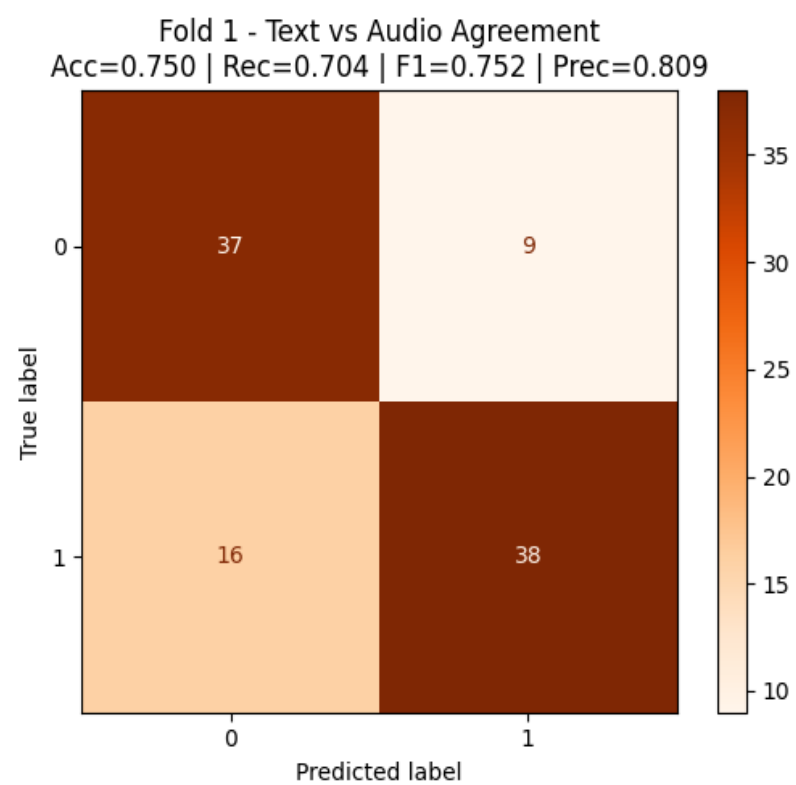
Text (English) Confusion Matrix



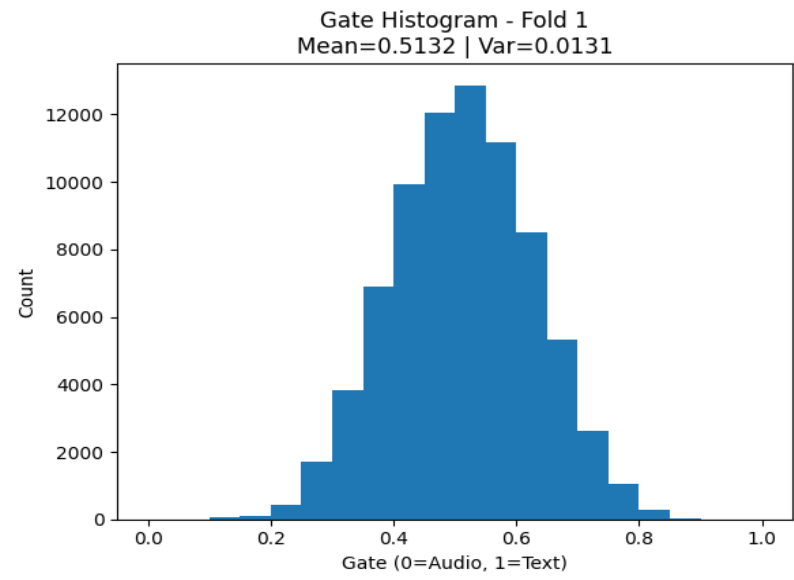
Text (Hindi) Confusion Matrix



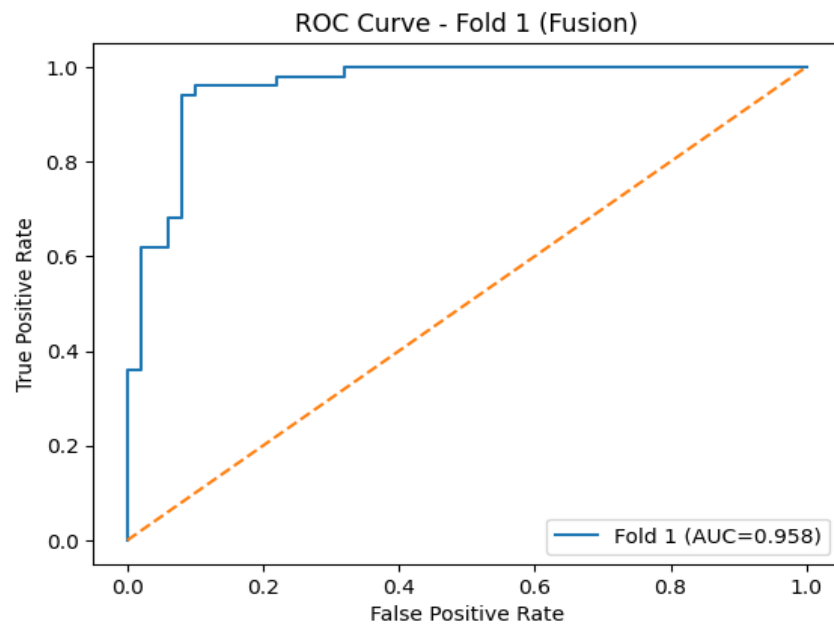
Text vs Audio Agreement



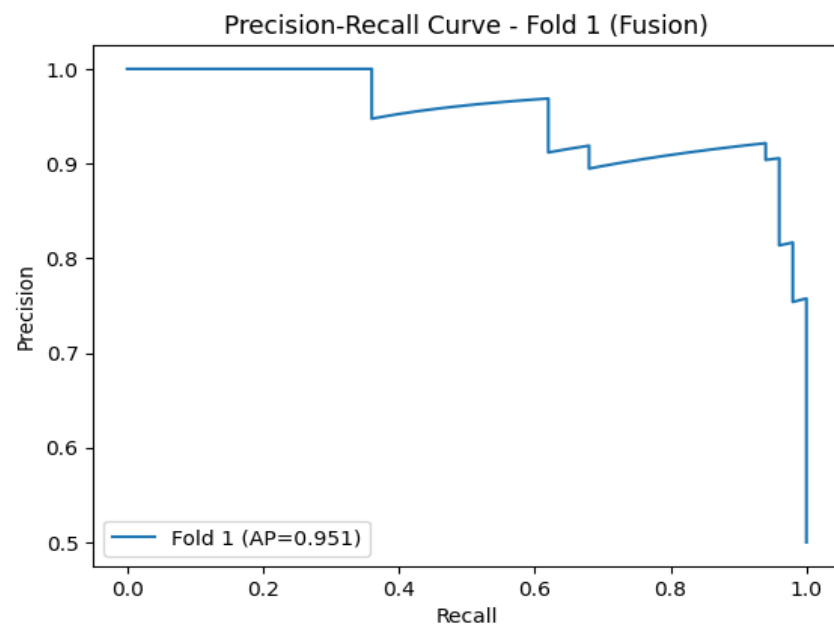
Fusion Gate Histogram



ROC Curve (Fusion)



Precision-Recall Curve (Fusion)

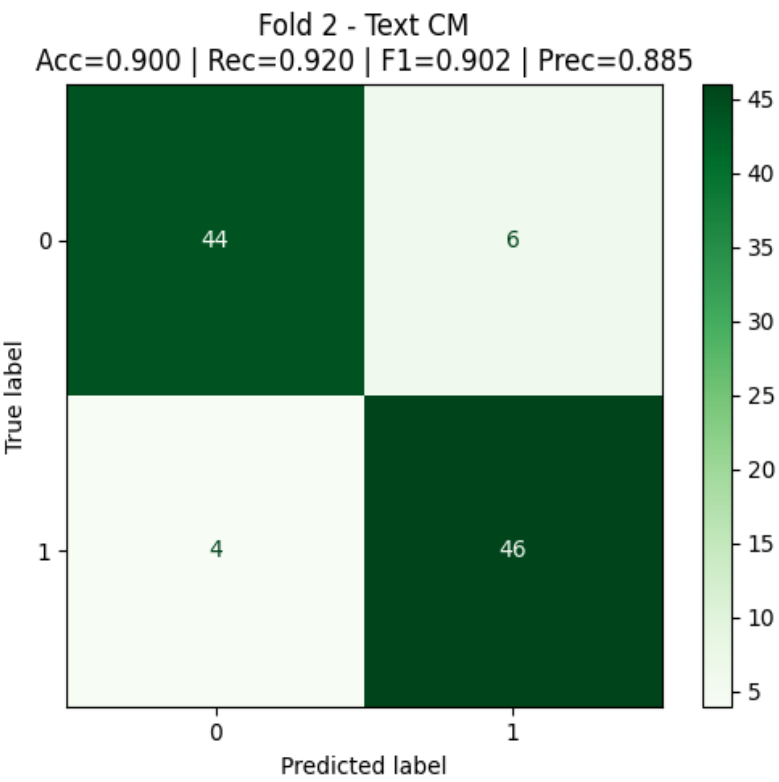


Metric	Text	Audio	Fusion
Accuracy	0.9000	0.8100	0.9100
Recall	0.8800	0.7333	0.9000
F1	0.8980	0.8224	0.9091
Precision	0.9167	0.9362	0.9184

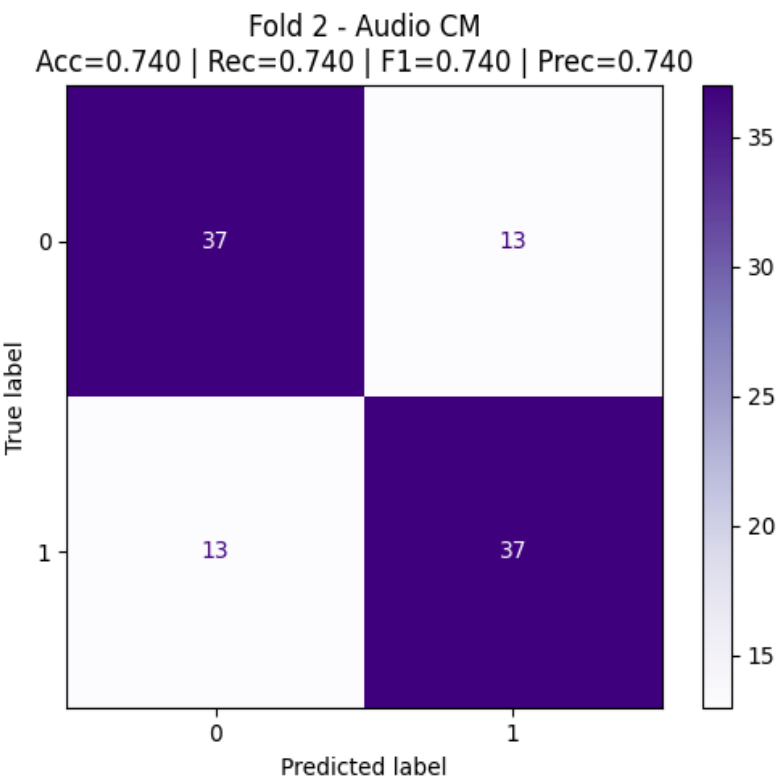
ROC AUC	-	-	0.9576
PR AUC	-	-	0.9511
Gate Mean	-	-	0.5132
Gate Var	-	-	0.0131

Fold 2 Results

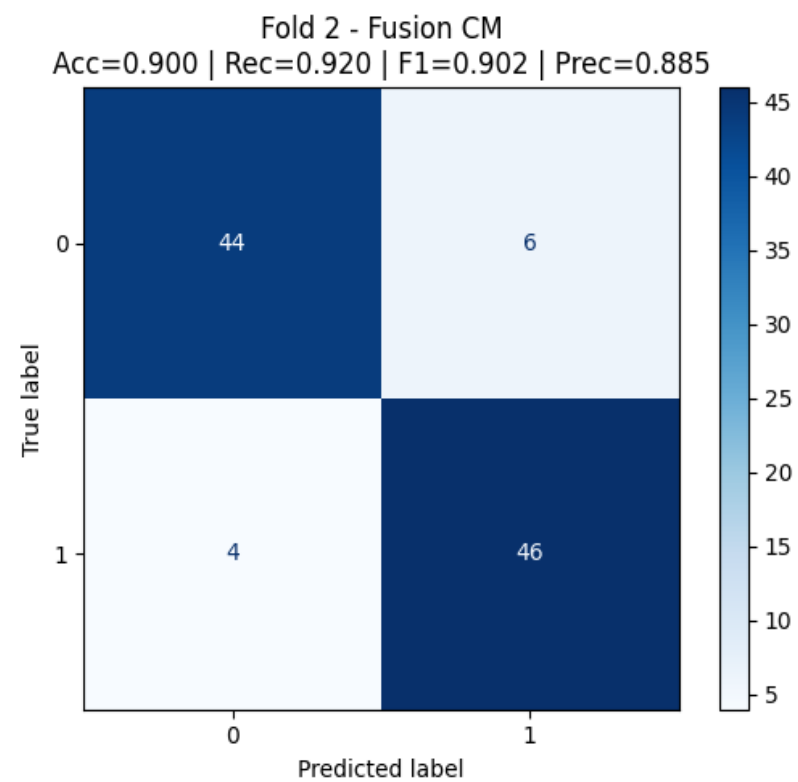
Text Confusion Matrix



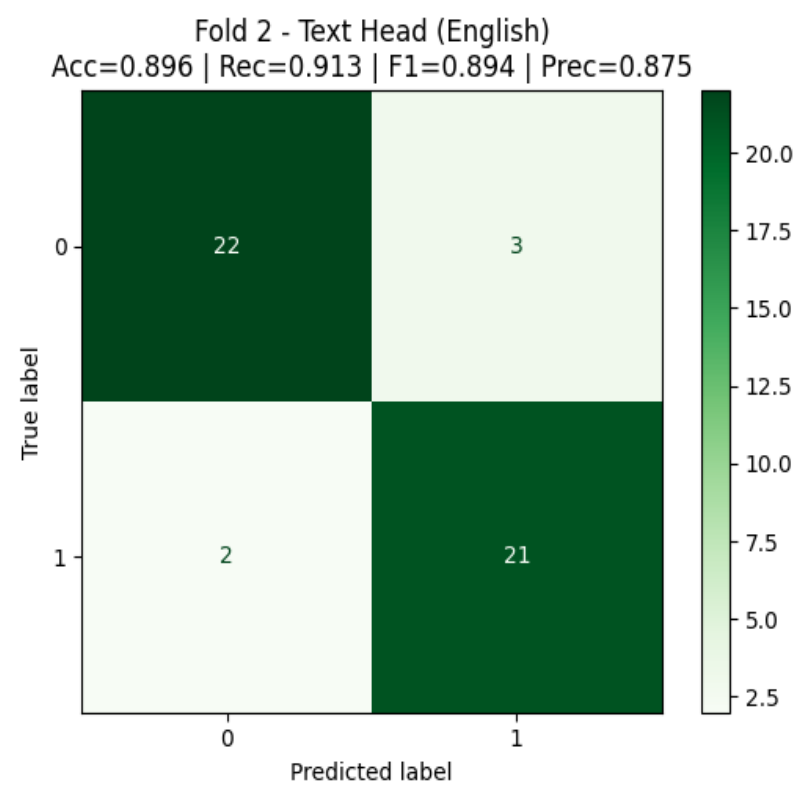
Audio Confusion Matrix



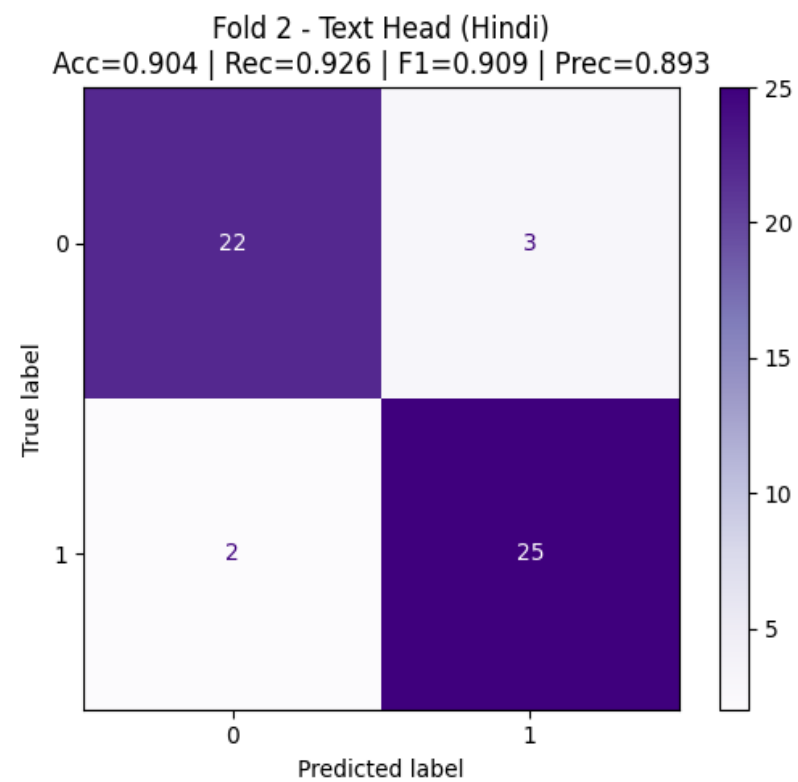
Fusion Confusion Matrix



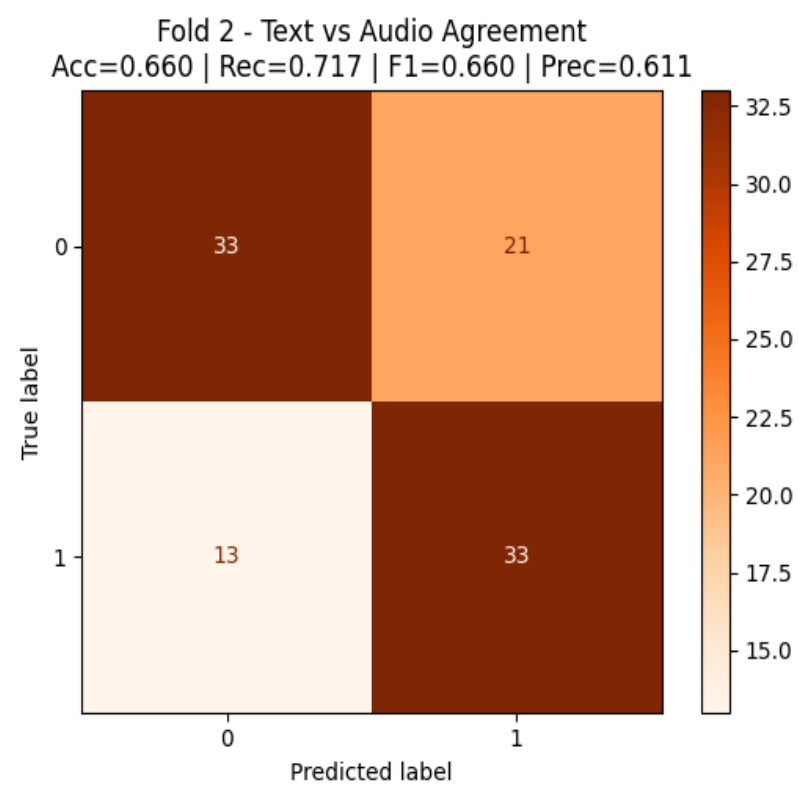
Text (English) Confusion Matrix



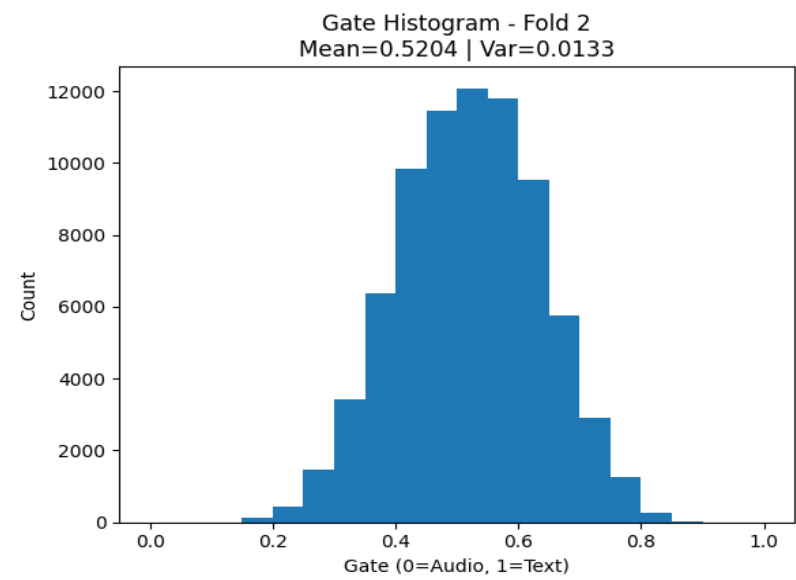
Text (Hindi) Confusion Matrix



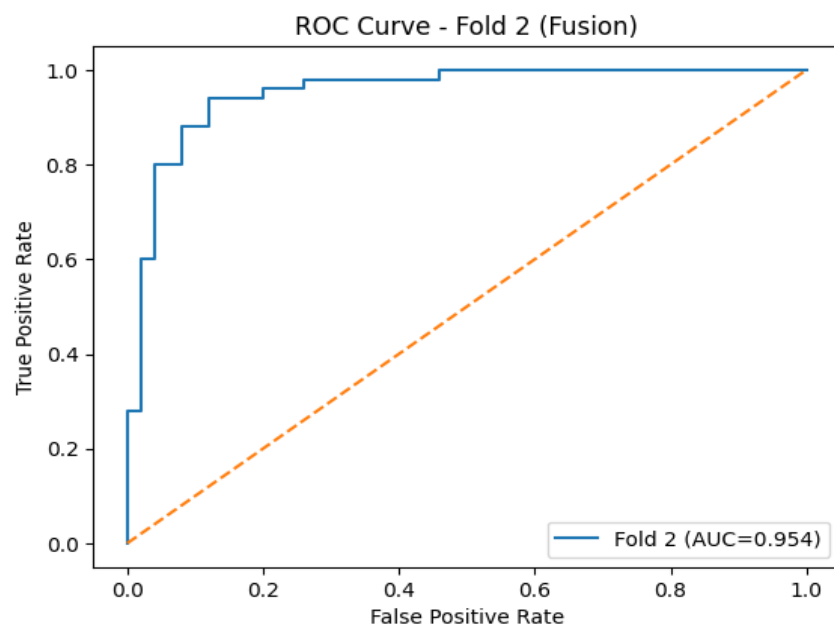
Text vs Audio Agreement



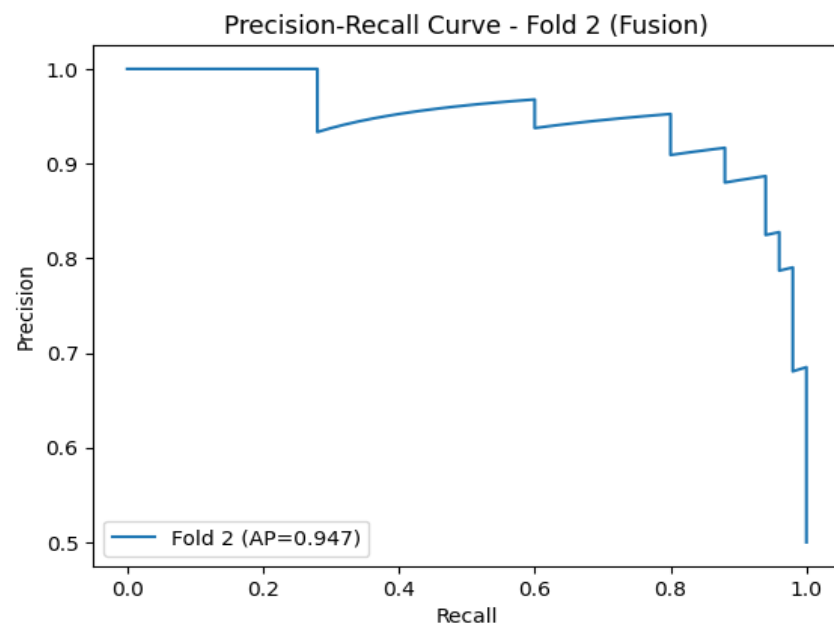
Fusion Gate Histogram



ROC Curve (Fusion)



Precision-Recall Curve (Fusion)

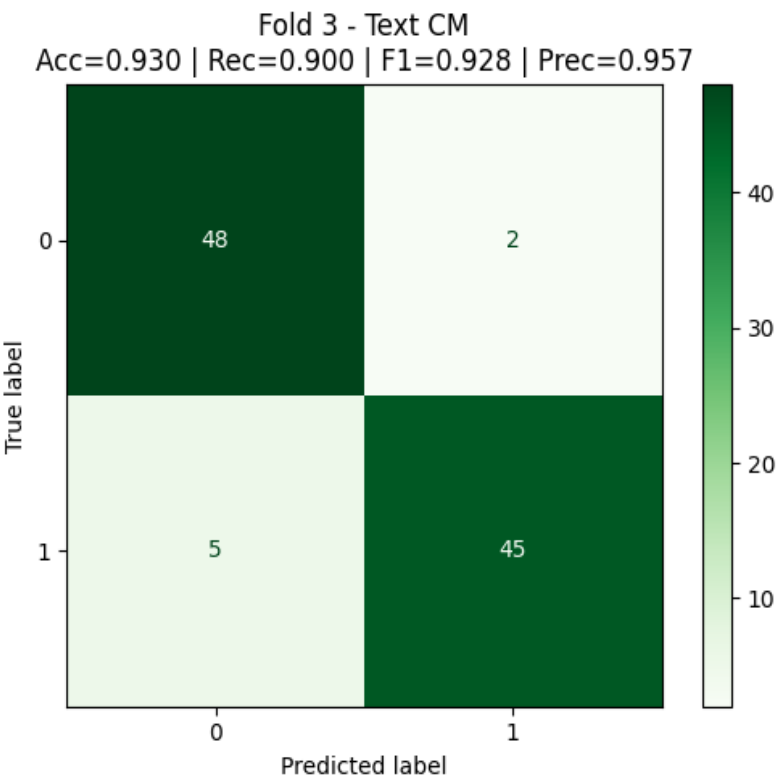


Metric	Text	Audio	Fusion
Accuracy	0.9000	0.7400	0.9000
Recall	0.9200	0.7400	0.9200
F1	0.9020	0.7400	0.9020
Precision	0.8846	0.7400	0.8846

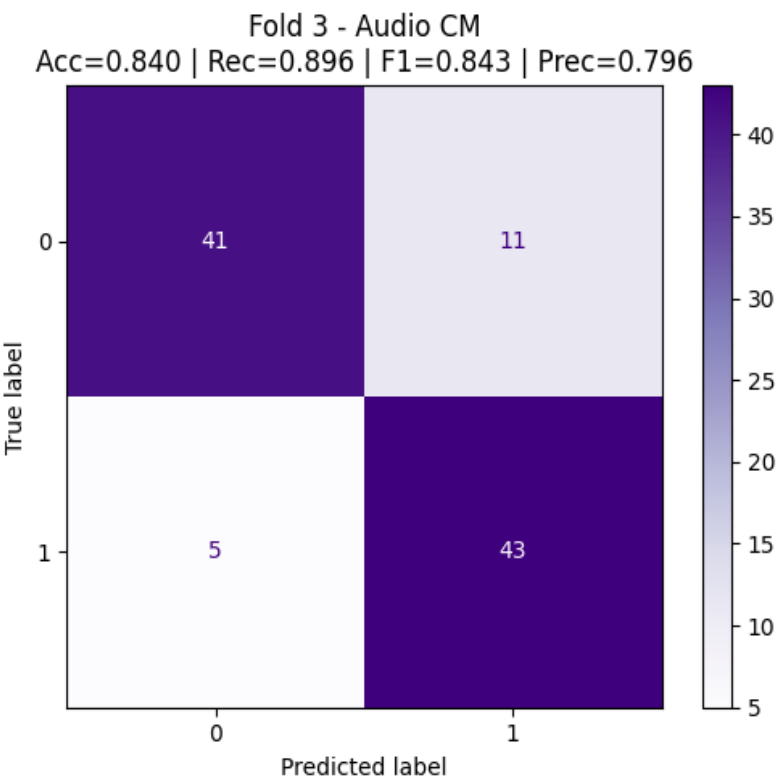
ROC AUC	-	-	0.9536
PR AUC	-	-	0.9473
Gate Mean	-	-	0.5204
Gate Var	-	-	0.0133

Fold 3 Results

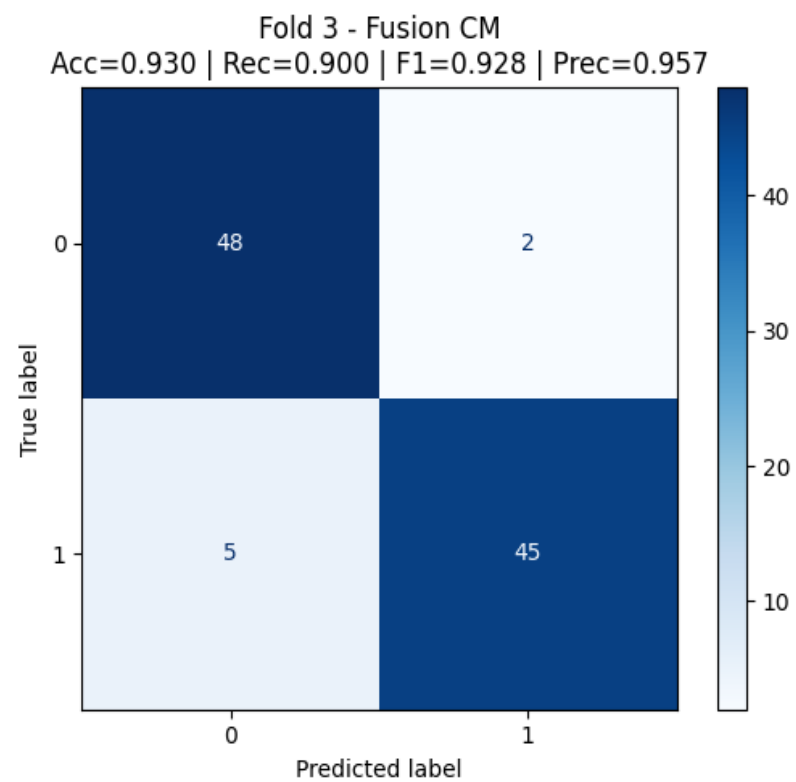
Text Confusion Matrix



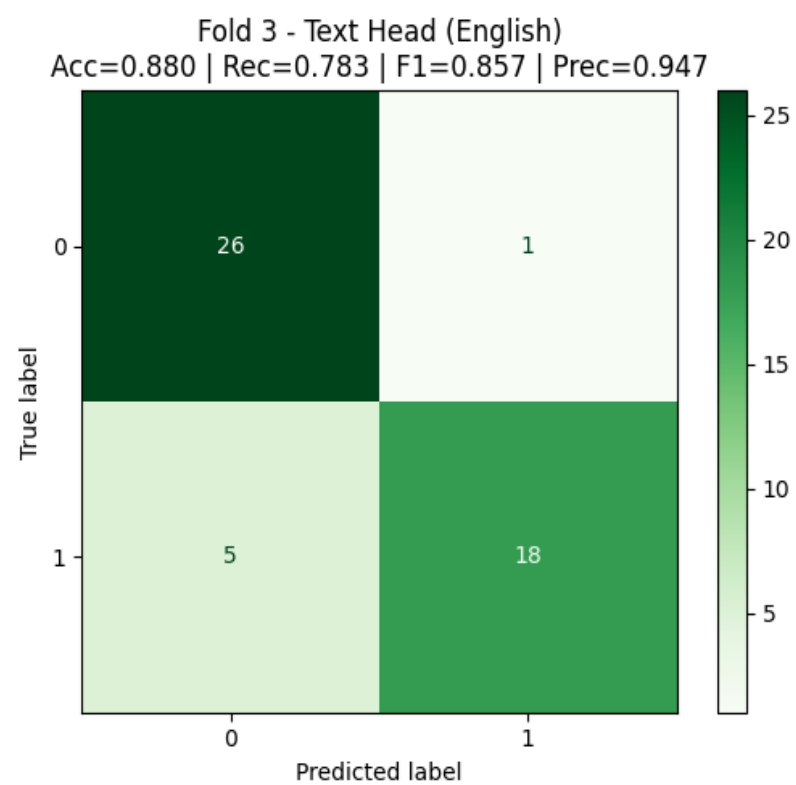
Audio Confusion Matrix



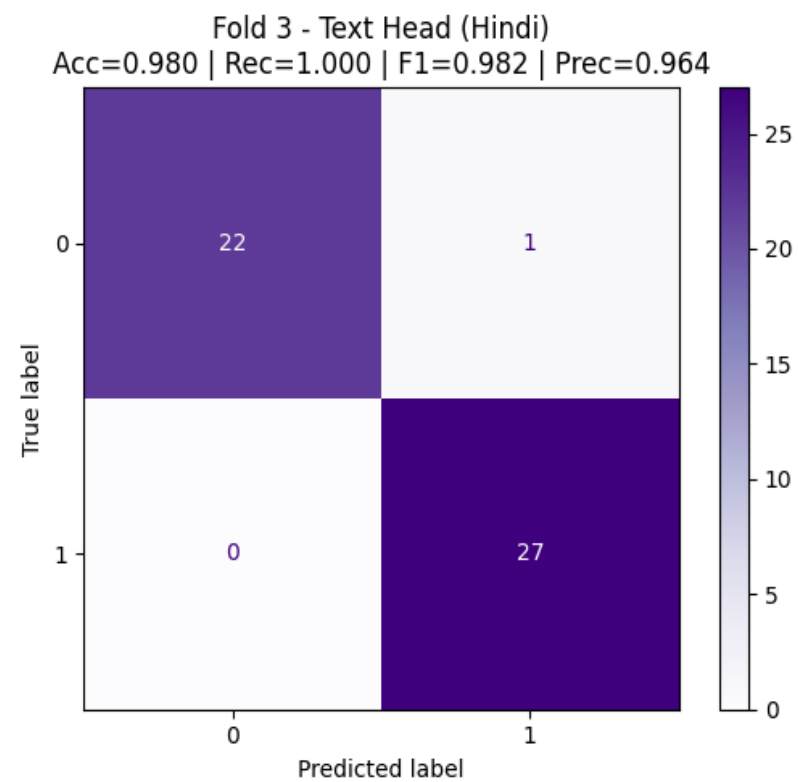
Fusion Confusion Matrix



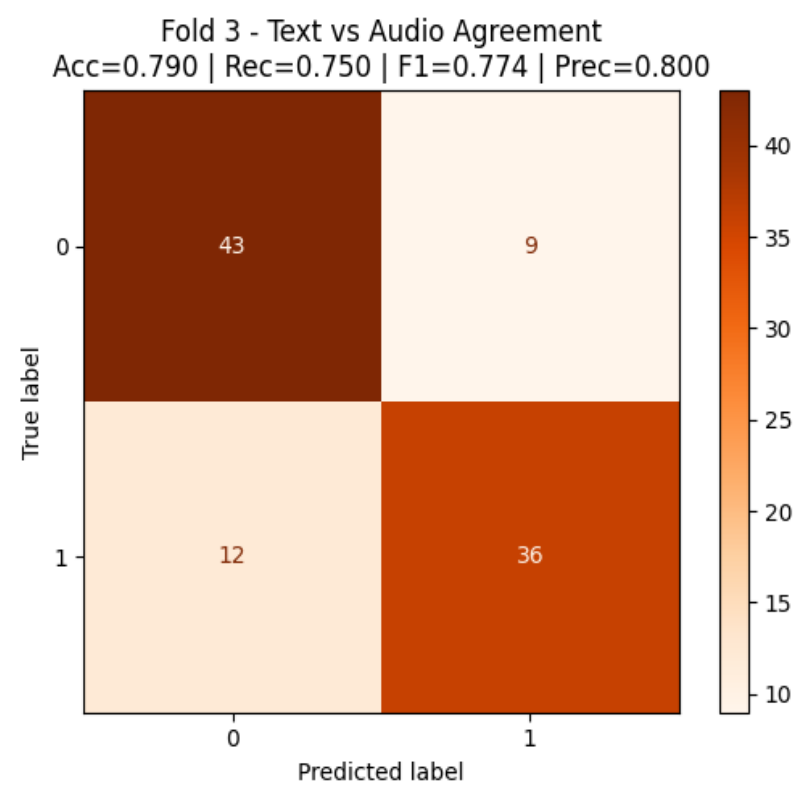
Text (English) Confusion Matrix



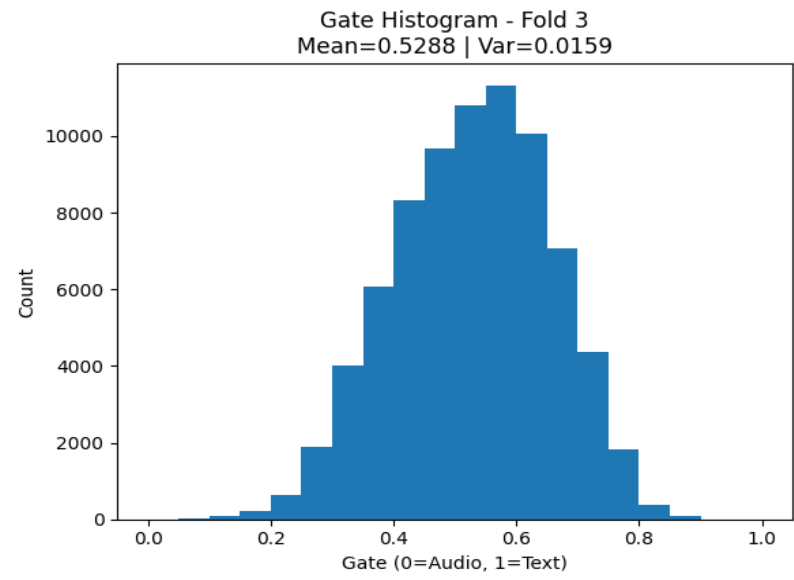
Text (Hindi) Confusion Matrix



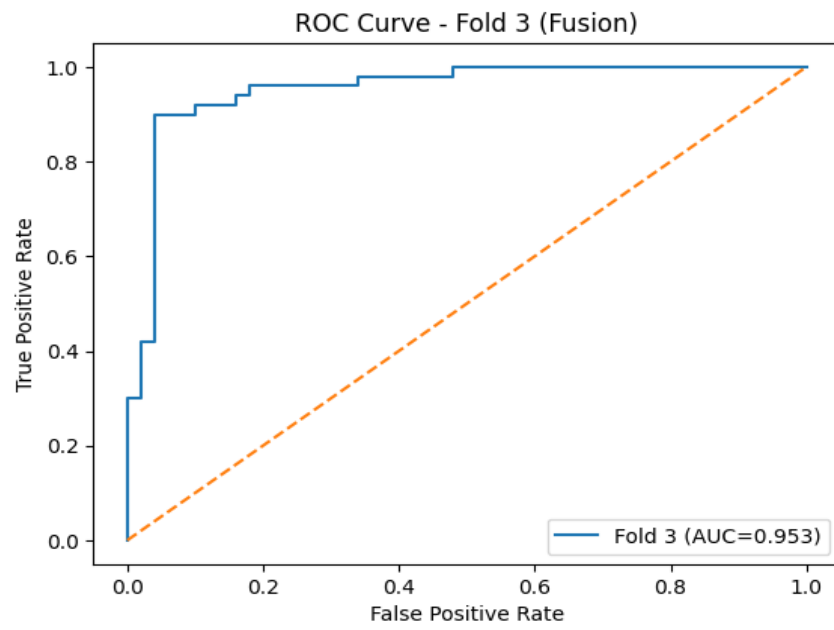
Text vs Audio Agreement



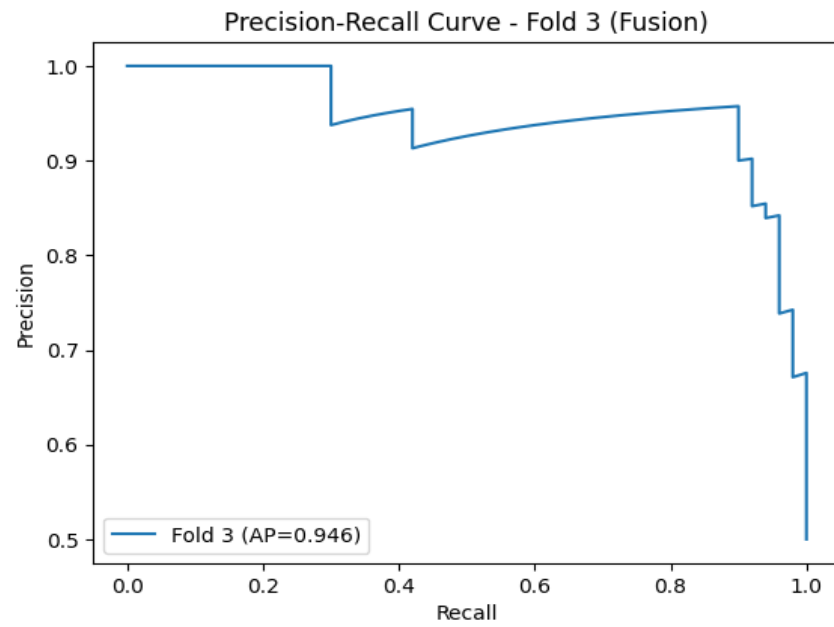
Fusion Gate Histogram



ROC Curve (Fusion)



Precision-Recall Curve (Fusion)

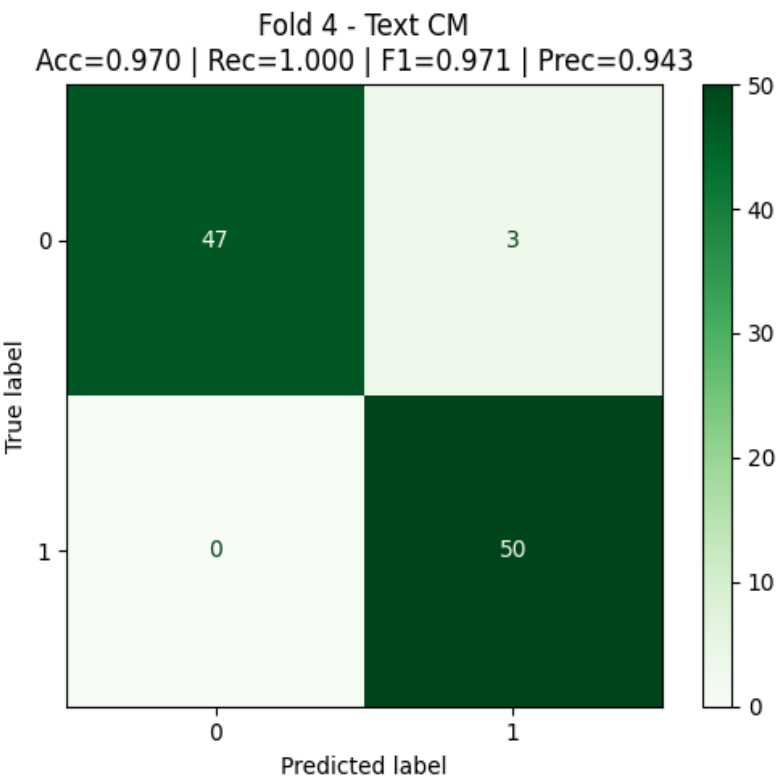


Metric	Text	Audio	Fusion
Accuracy	0.9300	0.8400	0.9300
Recall	0.9000	0.8958	0.9000
F1	0.9278	0.8431	0.9278
Precision	0.9574	0.7963	0.9574

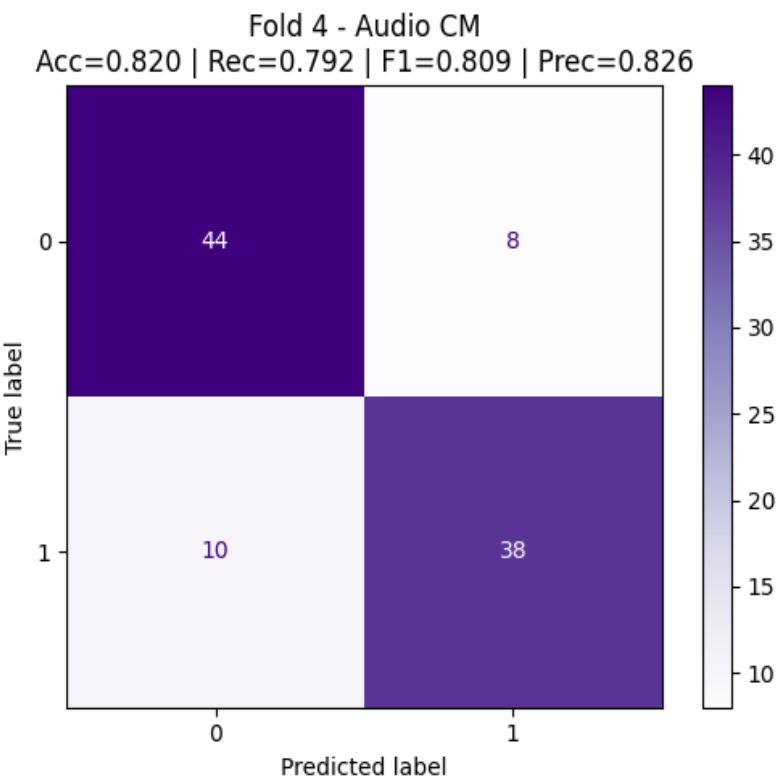
ROC AUC	-	-	0.9532
PR AUC	-	-	0.9460
Gate Mean	-	-	0.5288
Gate Var	-	-	0.0159

Fold 4 Results

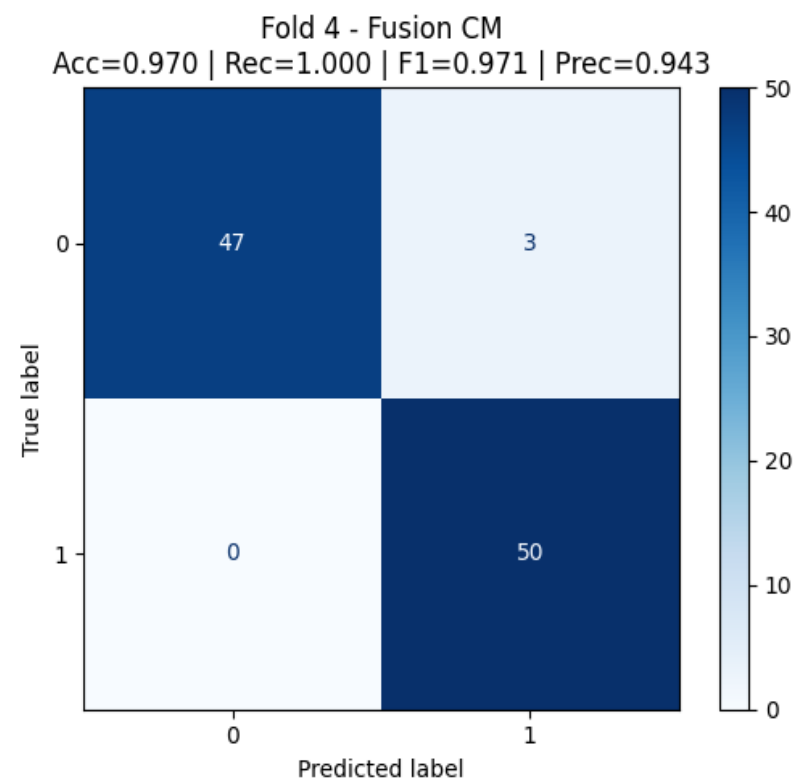
Text Confusion Matrix



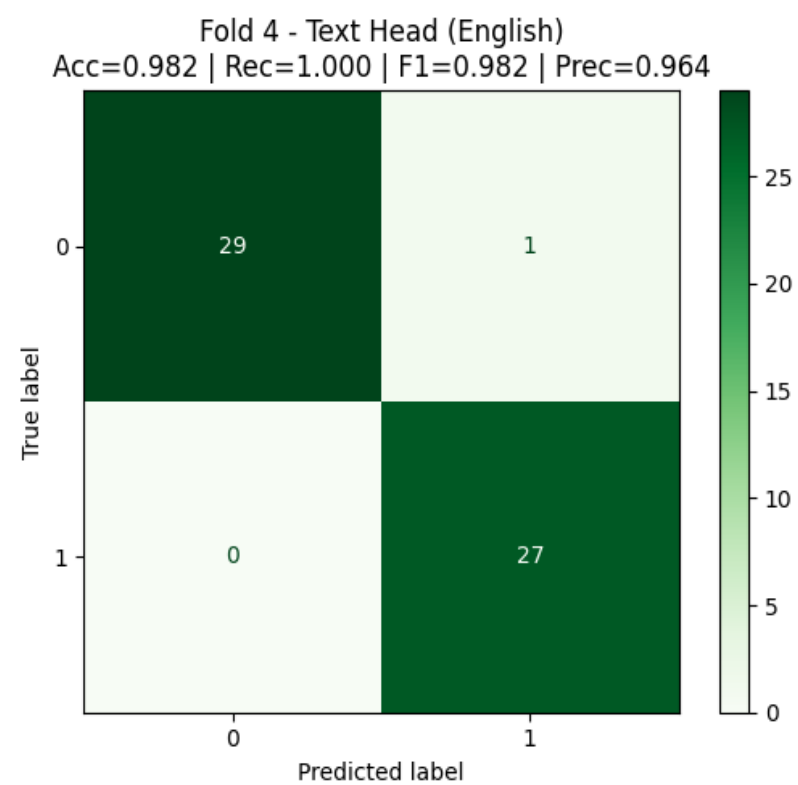
Audio Confusion Matrix



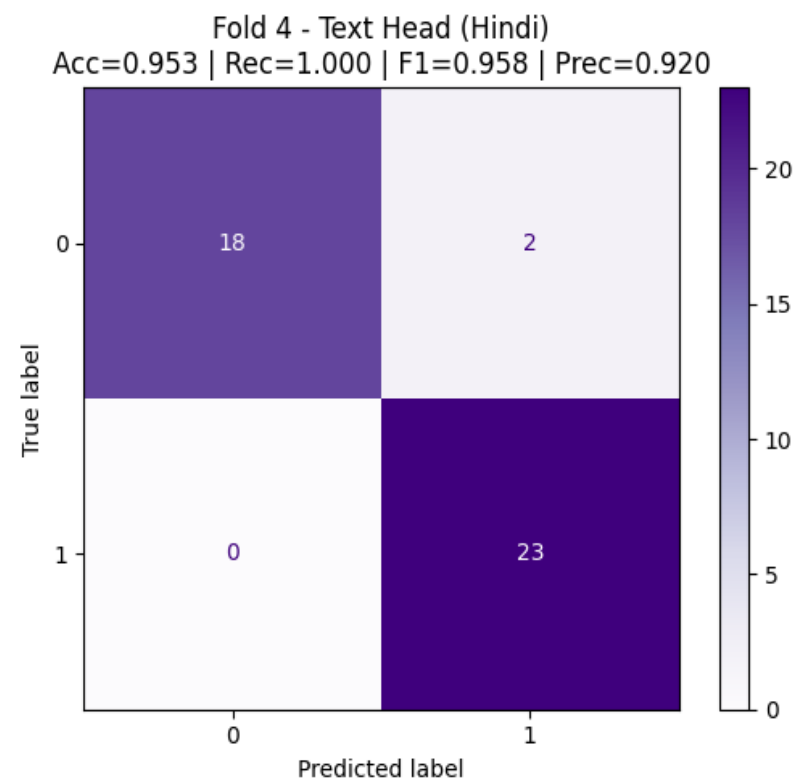
Fusion Confusion Matrix



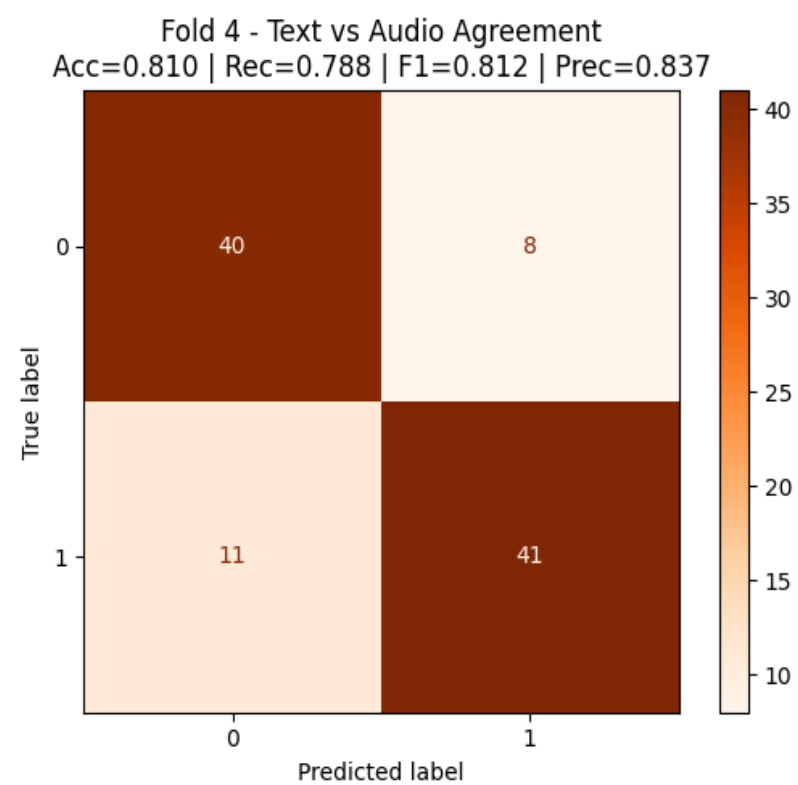
Text (English) Confusion Matrix



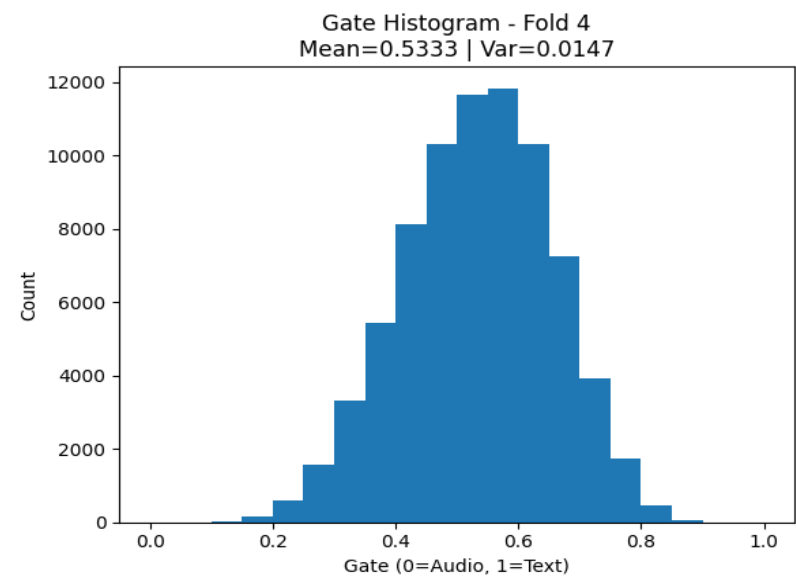
Text (Hindi) Confusion Matrix



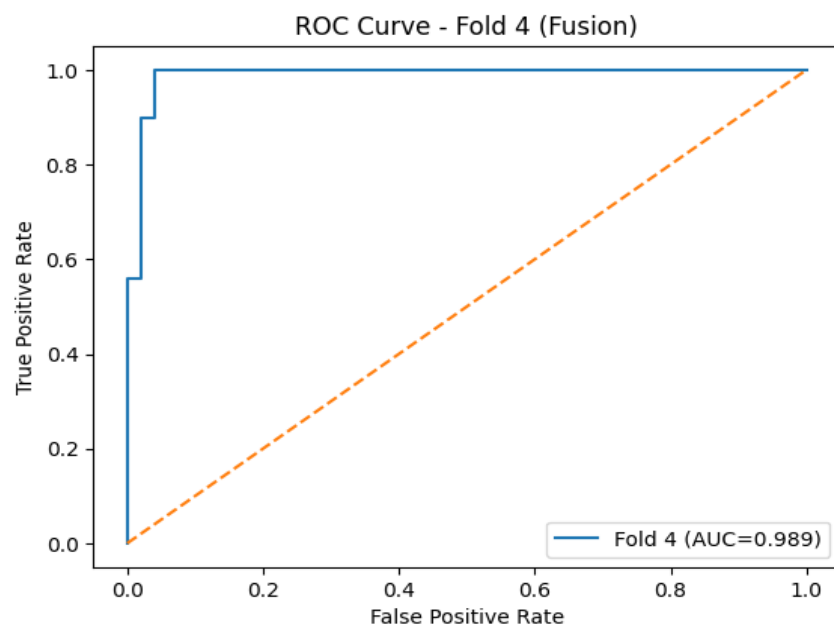
Text vs Audio Agreement



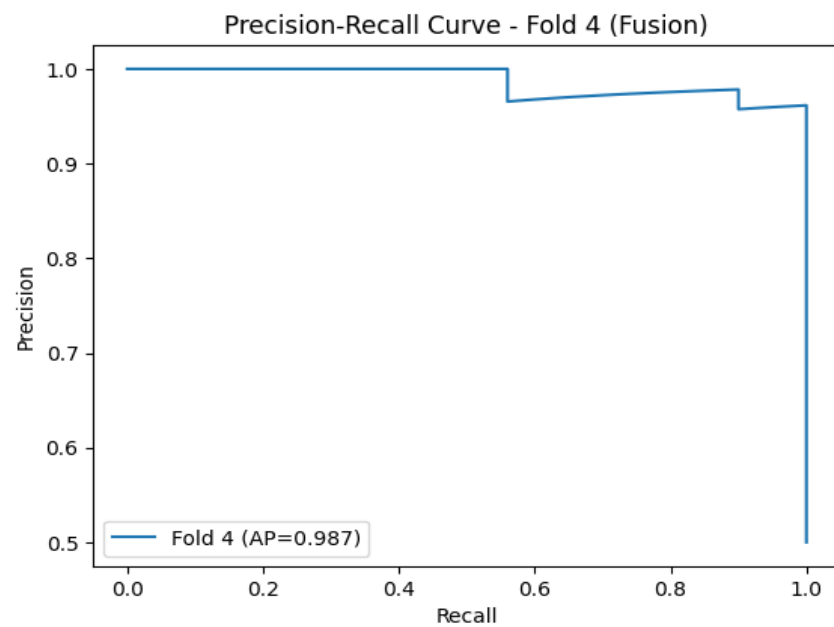
Fusion Gate Histogram



ROC Curve (Fusion)



Precision-Recall Curve (Fusion)

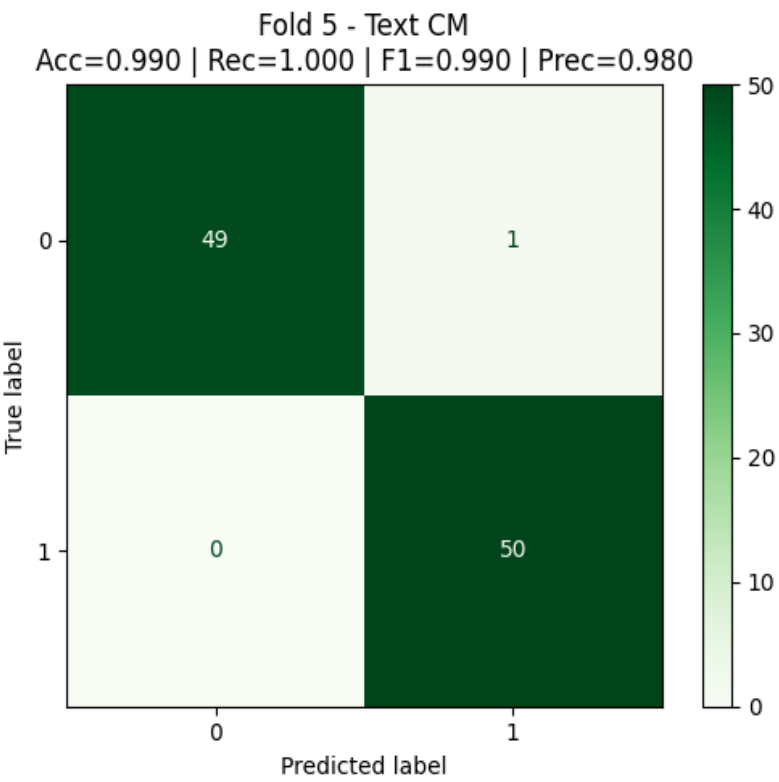


Metric	Text	Audio	Fusion
Accuracy	0.9700	0.8200	0.9700
Recall	1.0000	0.7917	1.0000
F1	0.9709	0.8085	0.9709
Precision	0.9434	0.8261	0.9434

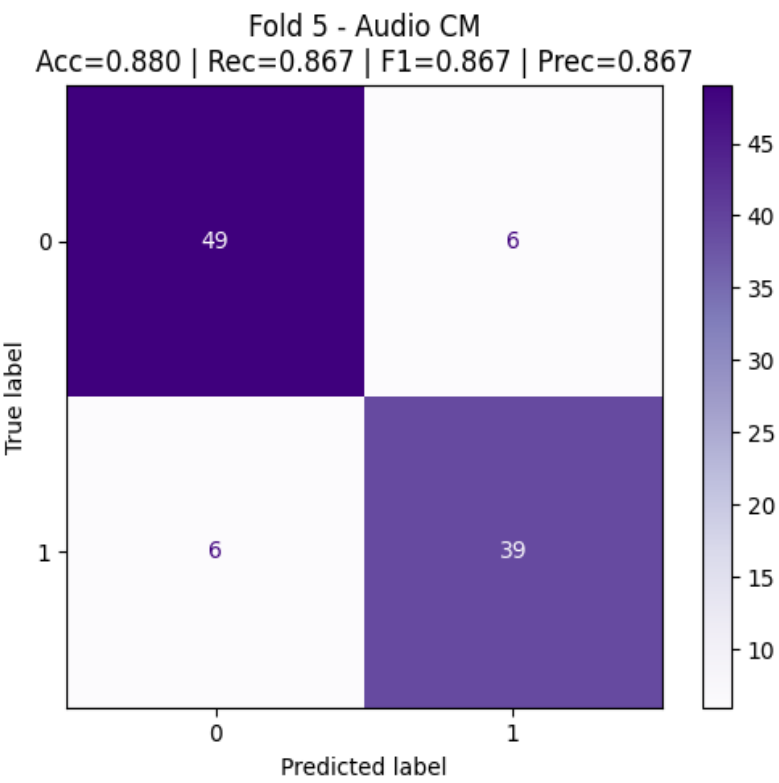
ROC AUC	-	-	0.9892
PR AUC	-	-	0.9869
Gate Mean	-	-	0.5333
Gate Var	-	-	0.0147

Fold 5 Results

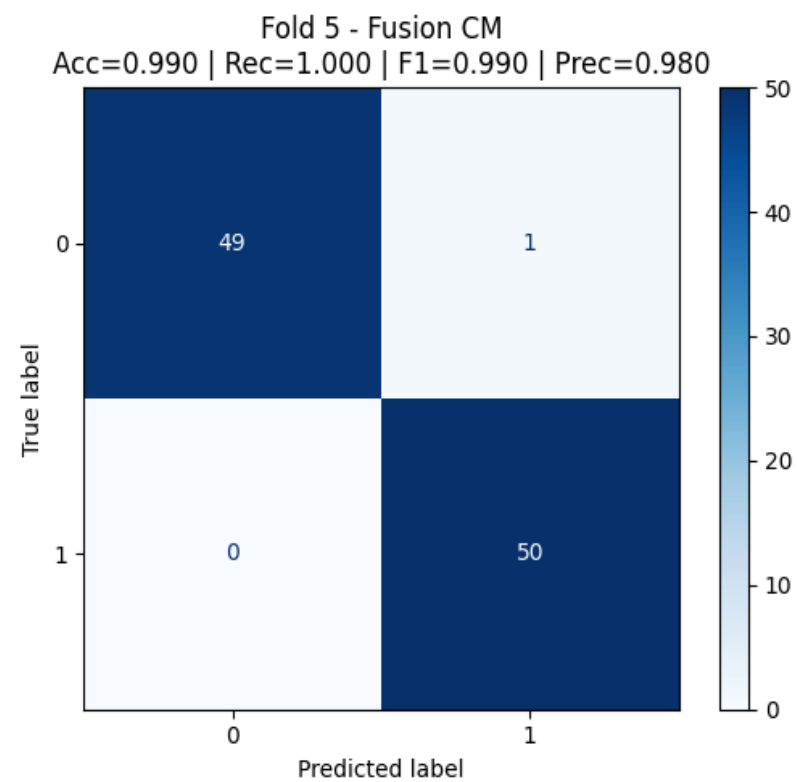
Text Confusion Matrix



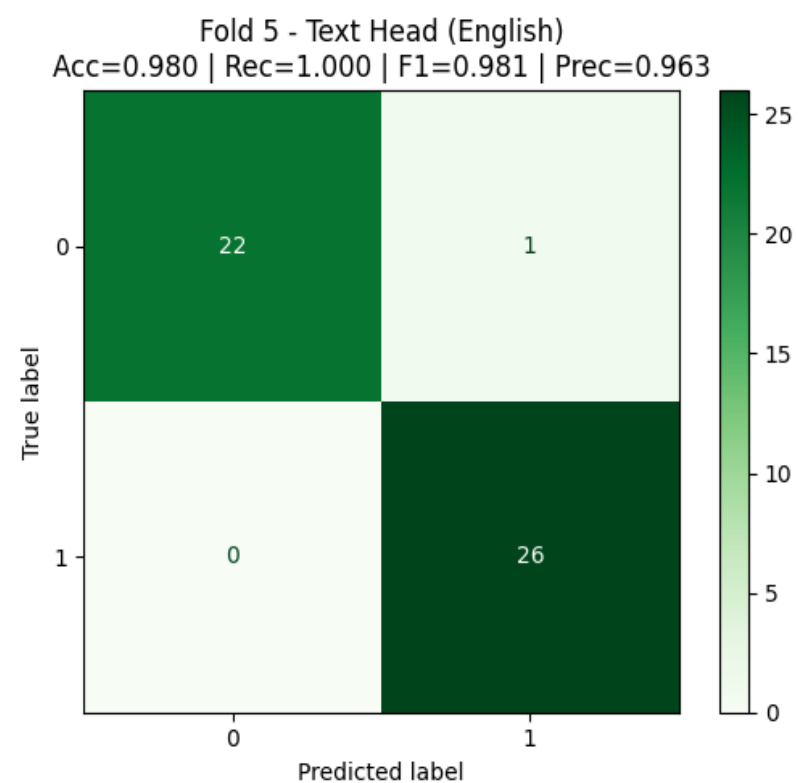
Audio Confusion Matrix



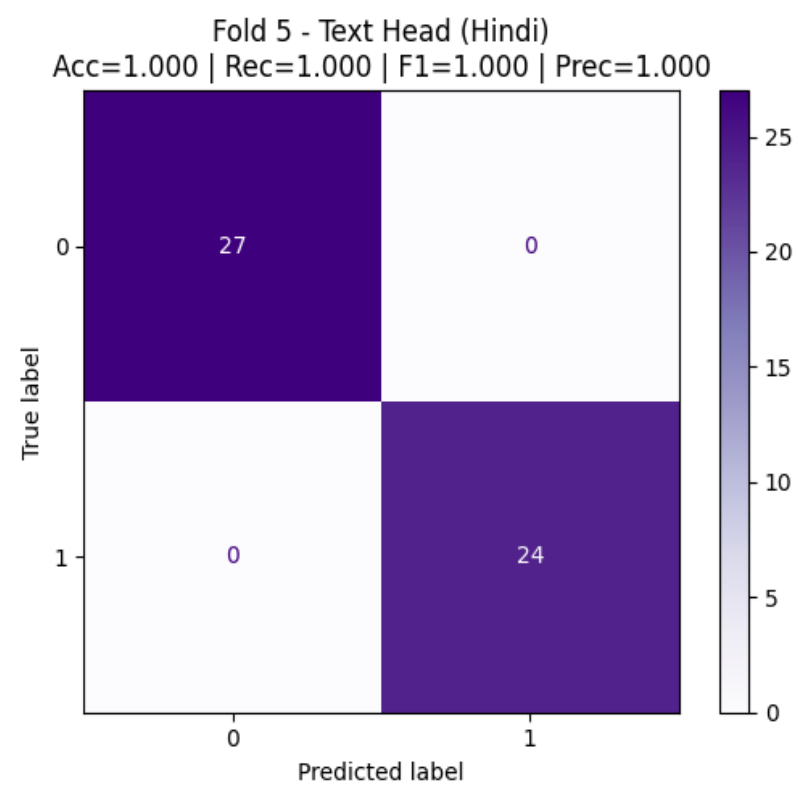
Fusion Confusion Matrix



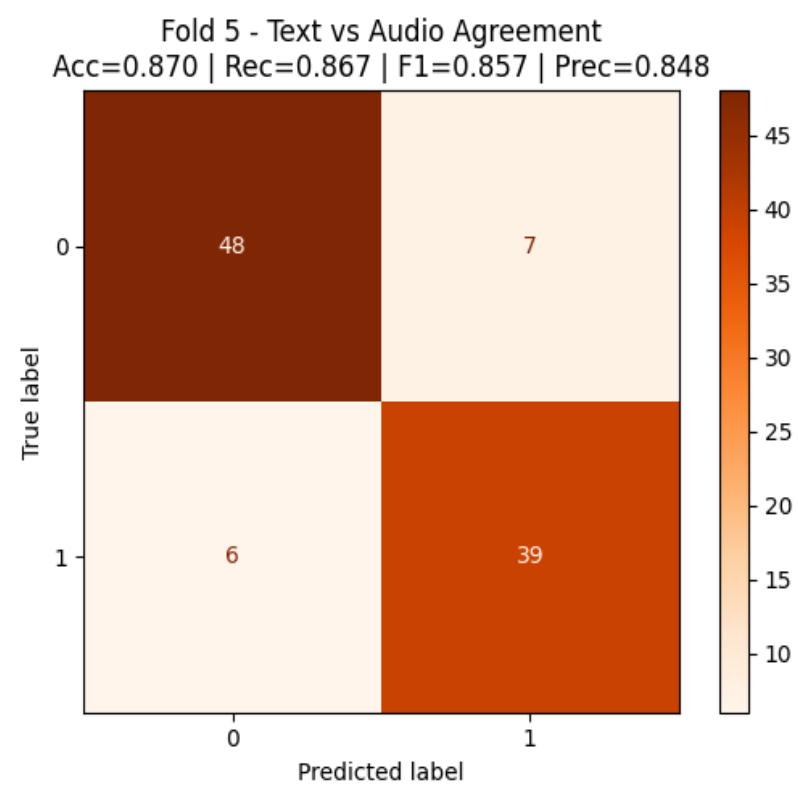
Text (English) Confusion Matrix



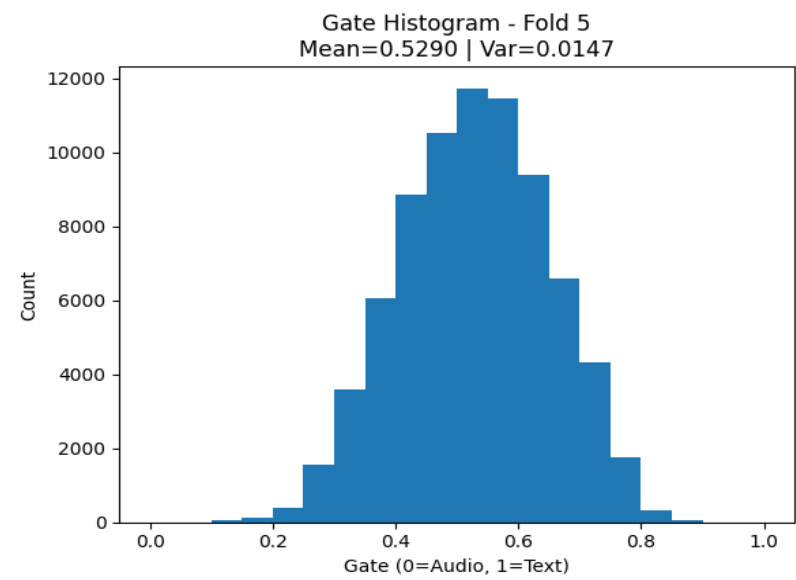
Text (Hindi) Confusion Matrix



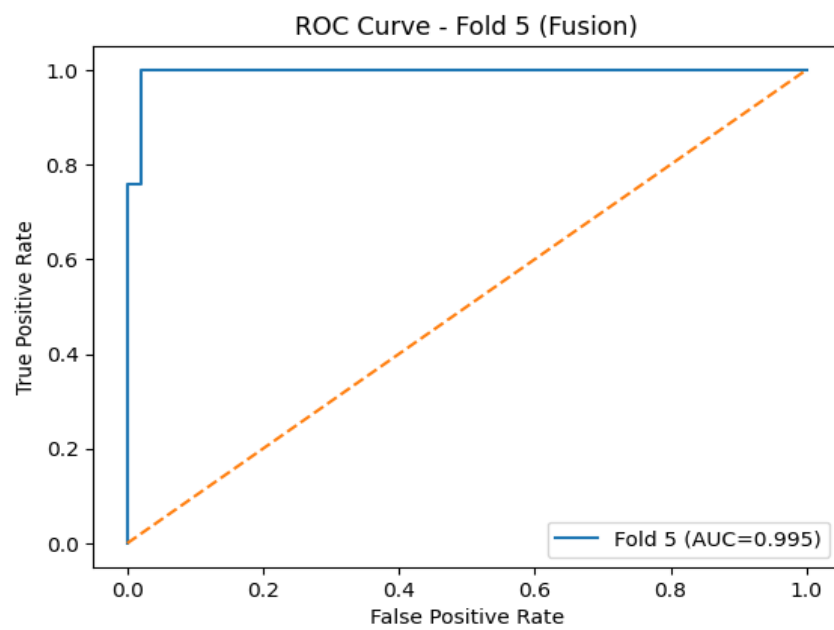
Text vs Audio Agreement



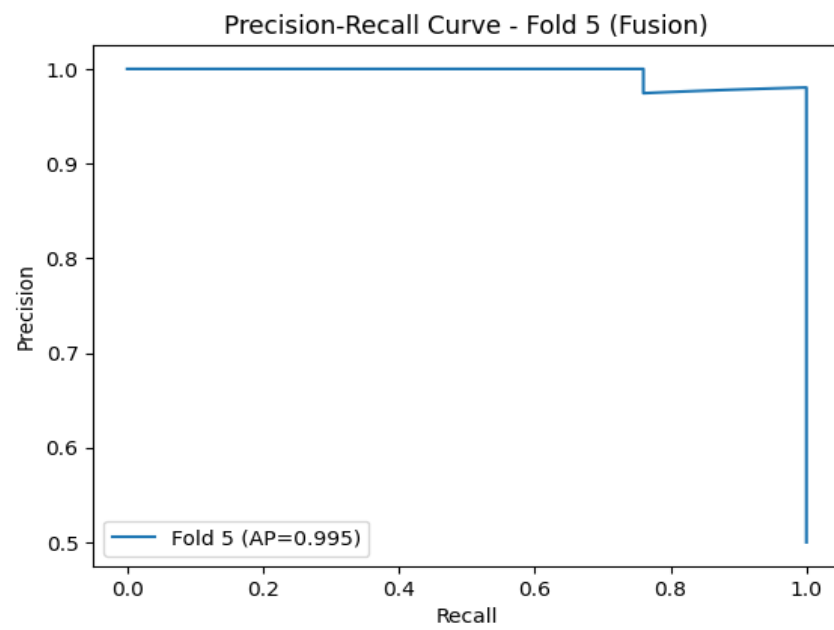
Fusion Gate Histogram



ROC Curve (Fusion)



Precision-Recall Curve (Fusion)



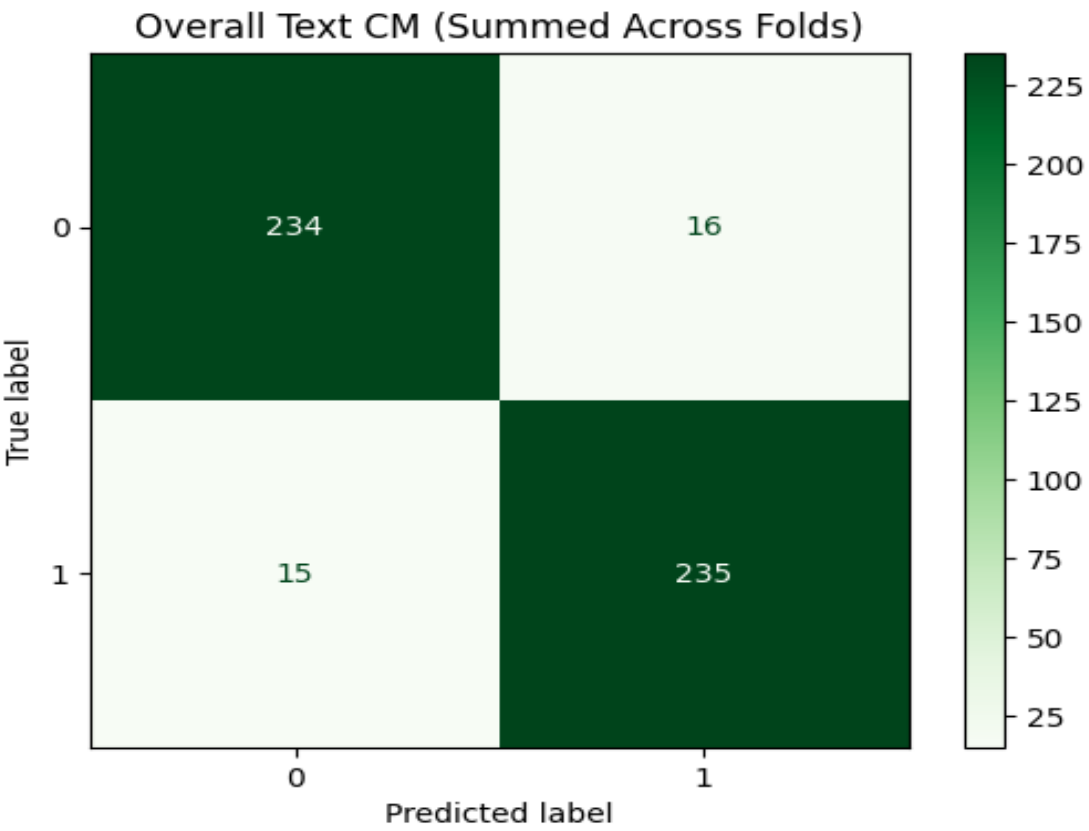
Metric	Text	Audio	Fusion
Accuracy	0.9900	0.8800	0.9900
Recall	1.0000	0.8667	1.0000
F1	0.9901	0.8667	0.9901
Precision	0.9804	0.8667	0.9804

ROC AUC	-	-	0.9952
PR AUC	-	-	0.9947
Gate Mean	-	-	0.5290
Gate Var	-	-	0.0147

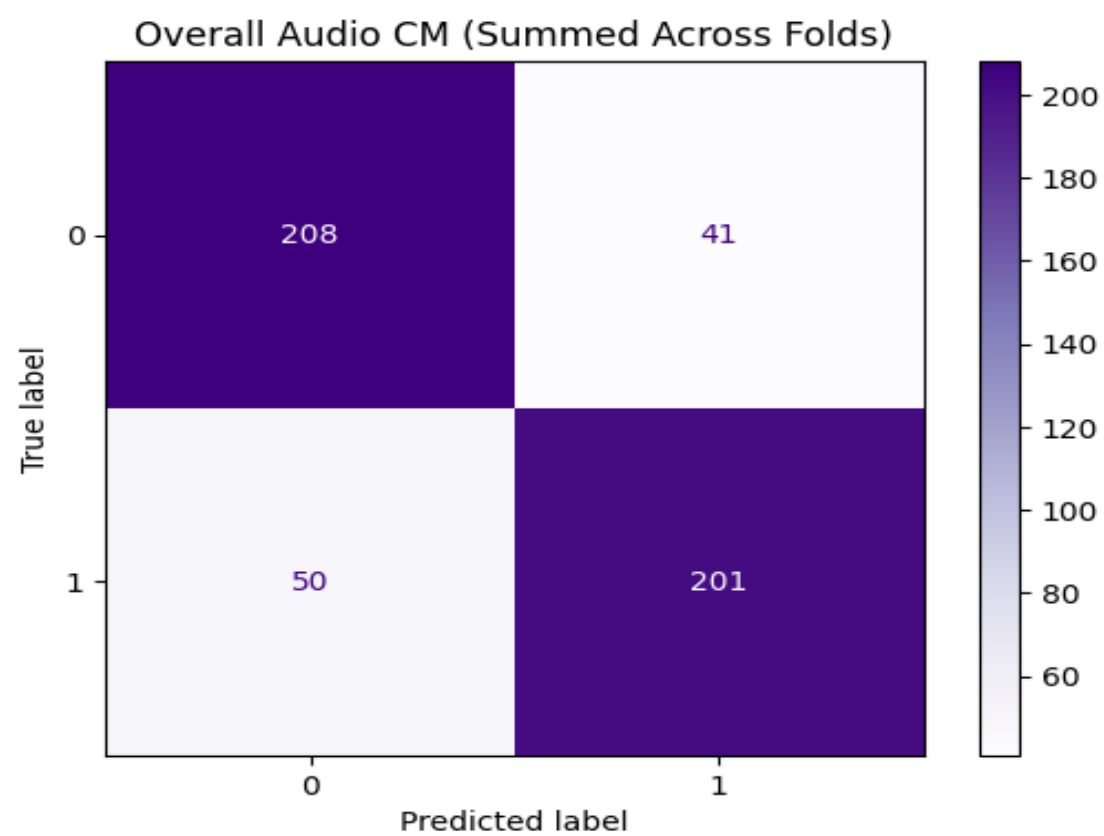
Aggregated Results Across Folds

Metric	Mean ± Std
Gate Mean	0.5250 ± 0.0072
Gate Variance	0.0143 ± 0.0011
Fusion Precision	0.9368 ± 0.0330
Fusion ROC AUC	0.9698 ± 0.0185
Fusion PR AUC	0.9652 ± 0.0211

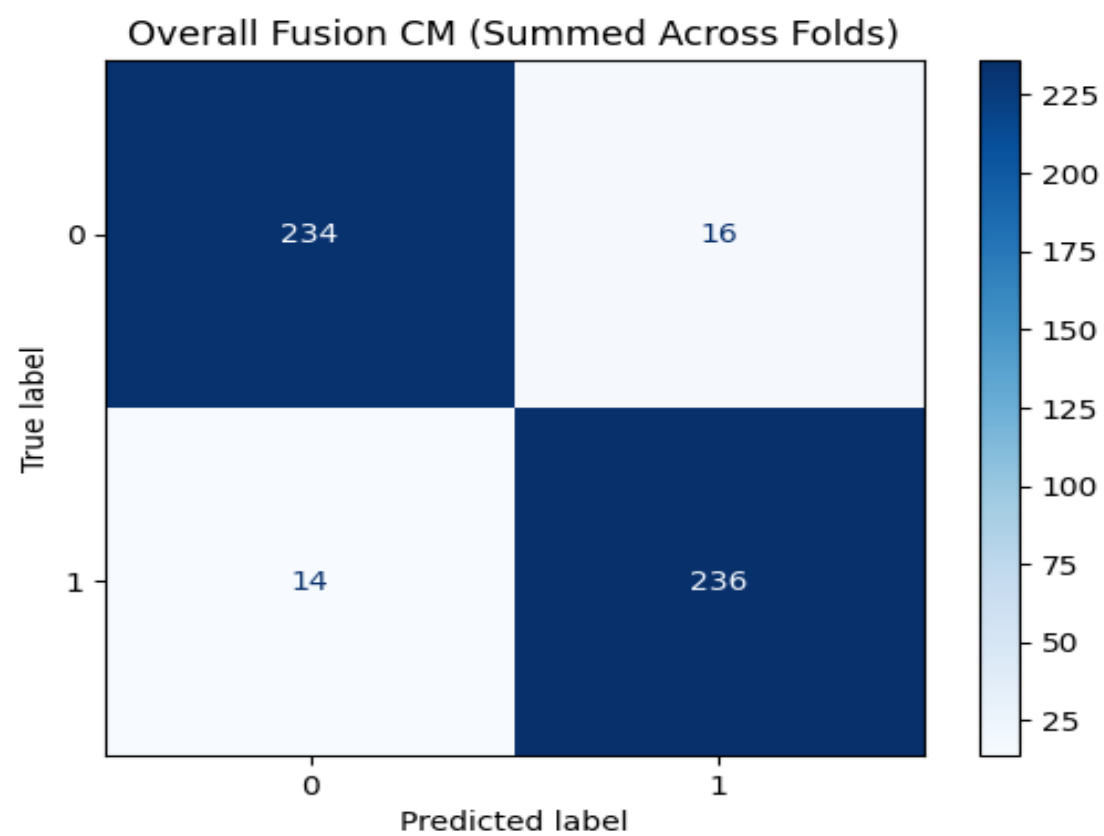
Overall Text CM



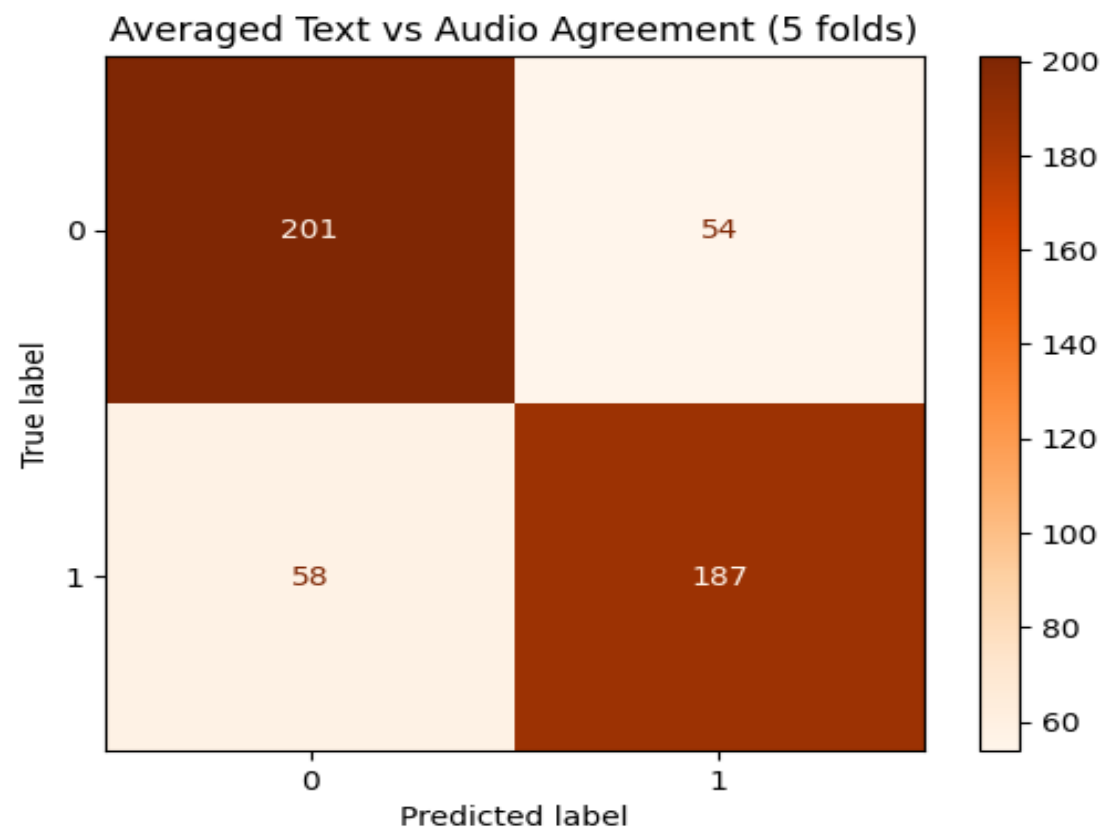
Overall Audio CM



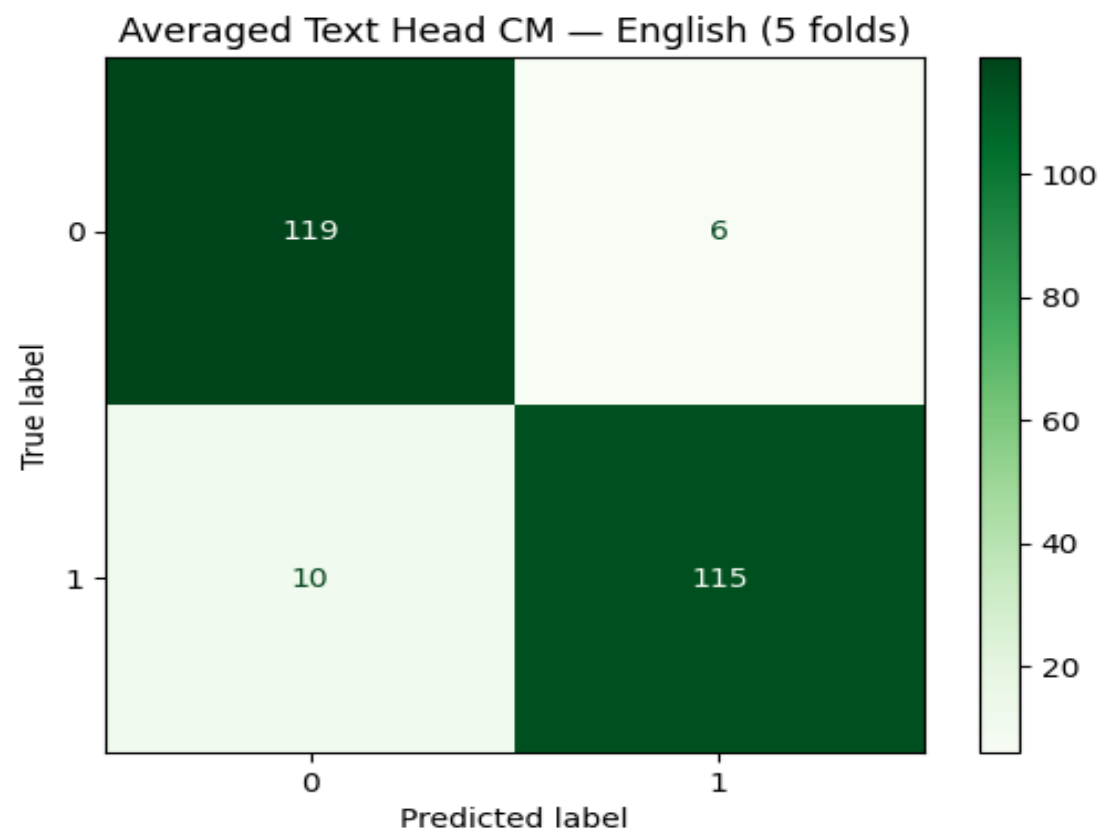
Overall Fusion CM



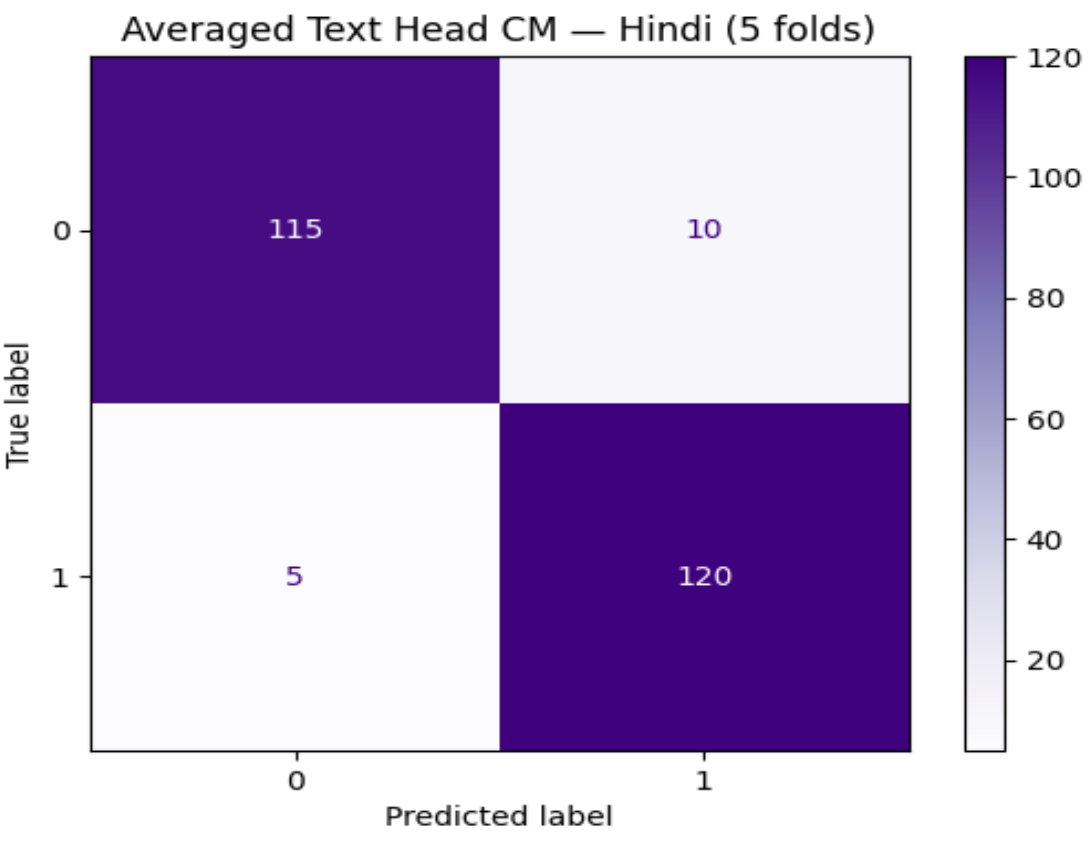
Averaged Text vs Audio Agreement (5 folds)



Averaged Text Head CM — English (5 folds)



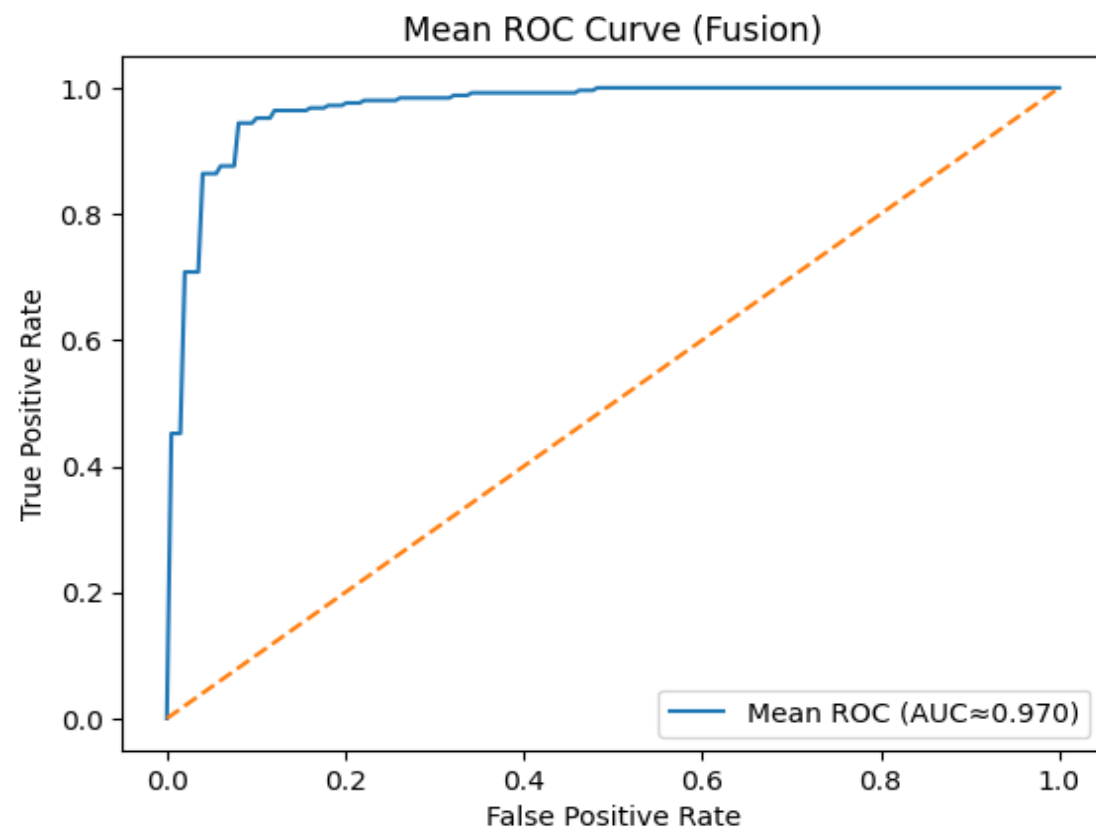
Averaged Text Head CM — Hindi (5 folds)



Averaged Global Metrics Table

Head	Accuracy ($\mu\pm\sigma$)	Recall ($\mu\pm\sigma$)	F1 ($\mu\pm\sigma$)	Precision ($\mu\pm\sigma$)
Text	0.938 $\pm$ 0.037	0.940 $\pm$ 0.051	0.938 $\pm$ 0.037	0.937 $\pm$ 0.033
Audio	0.818 $\pm$ 0.046	0.805 $\pm$ 0.066	0.816 $\pm$ 0.043	0.833 $\pm$ 0.066
Fusion	0.940 $\pm$ 0.035	0.944 $\pm$ 0.046	0.940 $\pm$ 0.035	0.937 $\pm$ 0.033

Mean ROC Curve (Across Folds)



Mean PR Curve (Across Folds)

